

Bias-free operational 200 GHz bandwidth photodetector toward 800 Gbps/ lane communication

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Abstract We proposed a bias-free operational 200-GHz photodetector assuming an integrated photoreceiver with a transimpedance amplifier toward the next-generation 800-Gbps per lane optical transceiver. The device performance of its high-baud-rate signal detection are discussed.

Introduction

With the increasing demand for artificial intelligence-based Internet in data centers, the data rate in short-range optical fiber communication has been expanding rapidly. No saturation trend has yet been observed with the increase in data traffic per year. A commercially available 400 Gbps optical transceiver [1] with 50 Gbaud PAM-4 $\times$ 4-lane was released in 2020. After two years, it was moved to an 800 Gbps transceiver [2] with 100 Gbaud PAM-4 $\times$ 4-lane. Although a 1.6 Tbps optical transceiver is mainly designed using 100 Gbaud PAM-4 $\times$ 8-lane [3], the configuration will be changed to 200 Gbaud PAM-4 $\times$ 4-lane [4-5] to produce a compact transceiver design in the near future. In this case, a minimum analog 3 dB bandwidth of 100–110 GHz will be required to consider the Nyquist frequency. When considering next-generation optical transceivers such as those with 3.2 and 6.4 Tbps, a higher baud rate of over 400 Gbaud will be required for the optical and electrical components. A high analog-frequency bandwidth of up to 200 GHz is crucial for realizing these transceivers.

In this study, we propose a bias-free operational high-speed photodetector (PD), assuming an integrated photoreceiver with a transimpedance amplifier (TIA) as an extremely high-data-rate optical transceiver toward 400 Gbaud (800 Gbps PAM-4) per lane communication. The design and fabrication of the PD and a demonstration of its high-baud-rate communication are discussed.

Bias-free operational PD with TIA circuit

In general, the photoreceiver in an optical transceiver consists of a high-speed PD and TIA to enhance the RF output signal. According to the commercially available 56 Gbaud TIA for 400 Gbps applications (100 Gbps-PAM-4 $\times$ 4 lanes) [6], the hybrid integration in these two devices is formed using bonding wires through an internal bias circuit in the TIA, as shown in Fig. 1(a). A simple implementation method between the PD and TIA using two bonding wires should allow stable and efficient RF output performance. However, the output performance may degrade when assuming a higher baud-rate

signal detection toward 400 Gbaud (800 Gbps-PAM-4) per lane, as shown in Fig. 1(a).

Fig. 2(a–c) show the simulated S21 O/E frequency response for (a) no two-bonding-wire inductance (L_w) as ideal case of Fig. 1(a), (b) two 50 pH bonding wires (two 50- μ m-long bonding wires) between the PD and TIA in Fig. 1(a), and (c) the PD anode connected to the signal line in the TIA using a 50 pH bonding wire, with the PD cathode connected to the ground (see Fig. 1(b)). We observed an approximately 50 GHz bandwidth difference between Fig. 2(b) and (c). It infers that a bias-free operational PD integrated with the TIA will offer a great advantage and potential for 400 Gbaud signal detection.

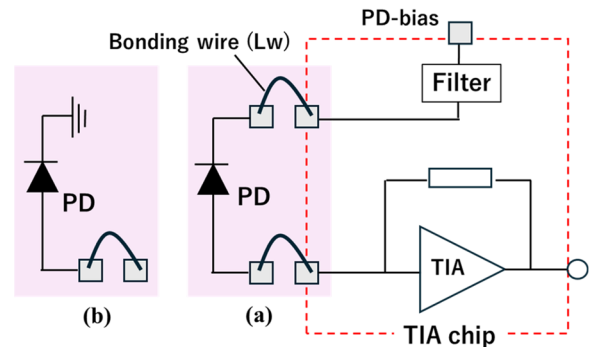


Fig. 1: Schematic circuit configuration of a PD integrated with a TIA, using (a) an internal bias feed from a TIA and (b) no bias feed to the PD

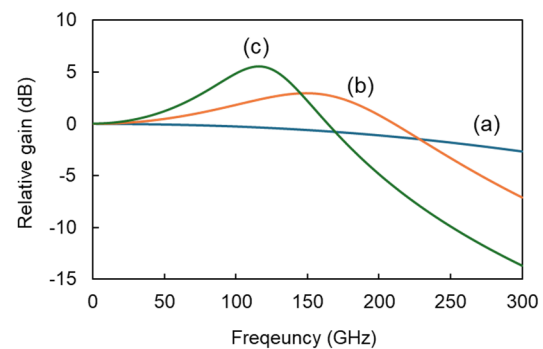


Fig. 2 Simulated S21 O/E response for (a) Fig. 1(a) with $L_w = 0$ pH, (b) Fig. 1(b) with $L_w = 50$ pH, and (c) Fig. 1(a) with $L_w = 50$ pH, with a PD capacitance of 15 fF

Design and fabrication

We designed an ultra-broadband 200 GHz PD by applying our previous work to a bias-free operational 100 GHz PD [7]. To achieve bias-free operation, the built-in potential in the pn-junction is important. We employed a uni-travelling carrier PD (UTC-PD) structure, which mainly consists of two important layers (an InGaAs absorption layer and an InP carrier collector layer) (see Fig. 3). Importantly, the carrier concentration (CC) and thickness of these two layers were carefully designed and optimized. Under the condition of a high p-doping level ($1 \times 10^{18} \text{ cm}^{-3}$) in the absorption layer, the built-in potential was dependent on the CC in the InP collector layer. A high built-in potential (0.71 eV) was obtained with a low CC InP layer in the energy band diagram.

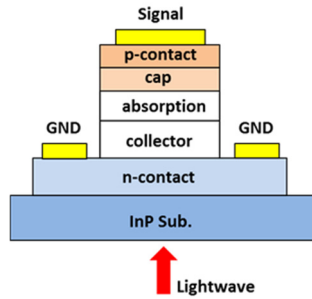


Fig. 3 Cross-sectional view of the bias-free operational PD

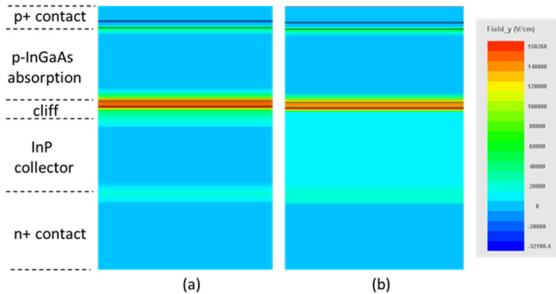


Fig. 4 Simulated internal electric field of PD at 0 V for (a) 10^{16} cm^{-3} range CC and (b) 10^{14} cm^{-3} range CC.

Moreover, the internal electric field distributions in UTC-PDs at 0 V were simulated with different CC conditions (a 10^{14} cm^{-3} range in this work and 10^{16} cm^{-3} range in previous work [8-9]). By comparing the electric fields under these two conditions, we found that an optimum electric field of 10 kV/cm could be obtained, even though no bias was applied. The highest electron drift velocity in the InP material was achieved at 10 kV/cm. In addition, the low CC collector layer could be attributed to a reduction in the junction capacitance. No electric field could be observed in the InP collector layer at the general CC (10^{16} cm^{-3} range) (see Fig. 4).

In the fabrication process, the 4- μm -diameter mesa structure was fabricated by a dry-etching

process and was covered with polyimide films. A ground-signal-ground (GSG) electrode was used to measure the high-frequency response. To extend the 3 dB bandwidth, the wire inductance on the signal line in the GSG electrode was considered, which affected the small peaking effect at high frequencies. The bottom-illuminated structure enhanced the responsivity. To achieve good impedance matching at high frequencies, a 50Ω thin-film resistor was integrated parallel to the PD element. The typical dark current of 10–20 nA was measured at -2 V.

Frequency response

Two types of measurement system were used for the S21 OE response at 220 GHz: a light-wave component analyzer in the range of 10 MHz–110 GHz, and an optical heterodyne system using two laser sources with an RF power meter in the range of 110–220 GHz. When comparing the frequency response at 0 V with that at 2 V, we found that the minimum degradation (20 GHz degradation) of the 3-dB bandwidth could be recognized between them (see Fig. 5(a) and (b)). A good frequency response of up to 200 GHz was successfully achieved, even when operating at 0 V. Good impedance matching can be observed in the inset of Fig. 5, and the S22 low return loss was found to be -10 to -30 dB at 70 GHz and -6 to -10 dB in the 70–110 GHz range. From a Nyquist frequency perspective, the high 3 dB bandwidth of 200 GHz suggests a potential to receive 400 Gbaud (800 Gbps PAM-4) signal detection at 0 V. The responsivity of the fabricated PD was as high as 0.1 A/W.

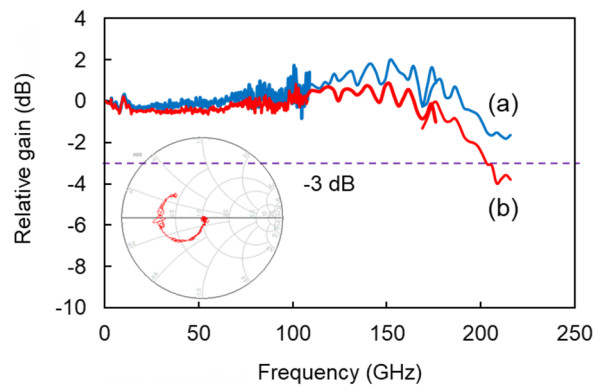


Fig. 5 Measurement results of O/E frequency response at (a) 2 V and (b) 0 V, with S22 Smith chart in inset

Potential for high-data-rate signal detection

For high baud rate communication potential on the fabricated PD, we demonstrated 126 Gbps (NRZ) to 252 Gbps (PAM-4) baud rate signal detection, which was the maximum baud rate available in our experimental setup. For the 126 Gbps detection, a pulse pattern generator

(PPG) without any pre-emphasis function on the PPG and without any equalization function on the oscilloscope was used. For the 252 Gbps PAM-4 signal detection, a weak equalization function (7-tap and 2-precursor) was used on the DSP in the oscilloscope.

First, while increasing the input photocurrent at 0 V, we measured the 100 Gbps NRZ output amplitude (eye amplitude) to confirm the output amplitude saturation behavior. When the photocurrent was increased to 2 mA, the output amplitude became proportional to the input photocurrent (see Fig. 6). Good linearity and an 80 mV output amplitude are sufficient to construct an integrated high-speed photoreceiver with a TIA. Owing to the fabrication error in the thin-film resistor, the amplitude was proportional to the 40 Ω termination line away from the 50 Ω termination line.

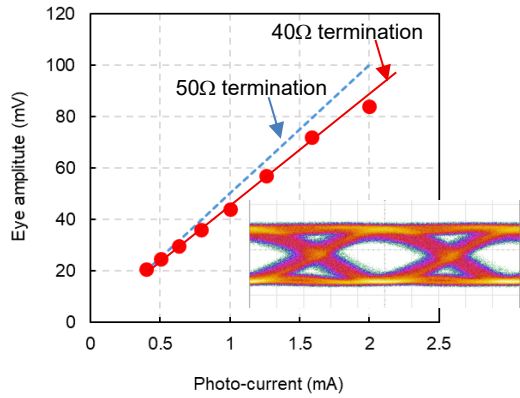


Fig. 6 Eye amplitude dependence of the input photocurrent at 100 Gbaud (NRZ) at 0 V, and observed eye-digram (inset)

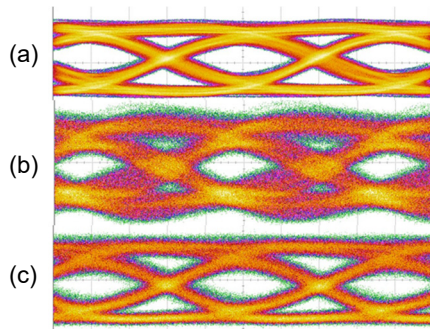


Fig. 7 Comparison of eye diagrams at 126 Gbaud (NRZ) (a) for the PPG electrical output in the transmitter, (b) using the commercially available 100 GHz PD at 2 V, and (c) using the bias-free operational 200 GHz PD at 0 V

Next, we compared the 126 Gbps (NRZ) eye diagrams of condition (a): original electrical PPG output with condition (b): PD output using the commercially available 100 GHz PD at 2 V, and condition (c): PD output with a 200 GHz bandwidth at 0 V in this work (see Fig. 7). A noisy eye diagram with a small eye opening and large time RMS jitter (1010 fs) was observed in

condition (b) compared with condition (a). In contrast, a very clear eye diagram with a large opening and low RMS jitter (607 fs) could be observed in condition (c), which was comparable to condition (a) with the same eye diagram quality. These results indicate that a higher-bandwidth PD improves the quality of the eye diagram.

Fig. 8(a) and (b) show the eye diagrams at 212 and 252 Gbps (PAM-4) for the fabricated PD at 0 V. With weak equalization of the DSP under the above conditions, clear eye diagrams could be successfully observed. The BER dependences of the data rate (180–252 Gbps) and input power on the PD are shown in Fig. 9(a) and (b), respectively. With +11 dBm input power without a TIA, 3.8×10^{-3} was sufficiently satisfied up to 252 Gbps.

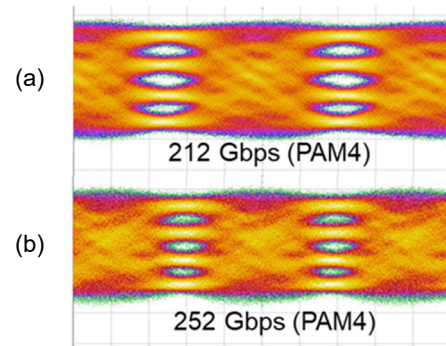


Fig. 8 Eye diagrams at (a) 212 Gbps and (b) 252 Gbps PAM-4 using the bias-free operational 200 GHz PD at 0 V

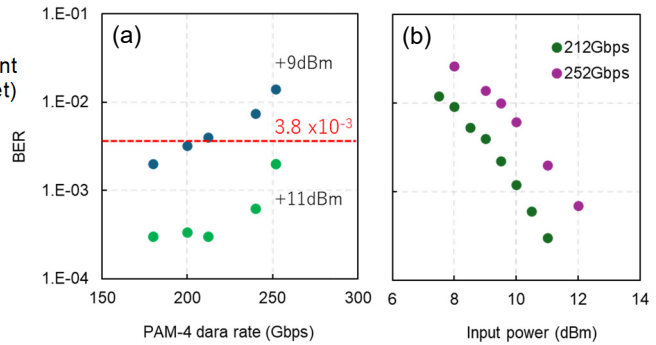


Fig. 9 BER dependence of the (a) data rate and (b) optical input power

Conclusions

We designed and fabricated a bias-free operational 200 GHz PD device assuming an integrated photoreceiver with a TIA toward the next-generation 400 Gbaud per lane optical transceiver. The proposed UTC-PD structure designed with a low-CC InP collector layer offered a high 3 dB bandwidth of up to 200 GHz at 0 V. High-quality eye diagrams of up to 252 Gbps suggested significant potential for next-generation extreme high-baud-rate communication.

Acknowledgements

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